

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1630

Xit 69 kV

November 18, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1630. The Interconnection Customer proposes a 110.4 MW solar photovoltaic facility interconnecting at the Xit 69 kV substation in the North Carolina area. The request was received on December 23, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 603 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$37,115,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Trowbridge 138-69 kV TX
Contingency: Home – Bel Air Substation 138.0 kV line
Loading: 140.9% (Rating: 92 MVA)
DFAX: 26.2%
Pre-project loading: 109.5%
2. Facility: Trowbridge 138-69 kV Auto Unit 2
Contingency: Bel Air – Home 138.0 kV
Loading: 135.6% (Rating: 96 MVA)
DFAX: 26.2%
Pre-project loading: 105.4%
3. Facility: Jajome 3403 138-69 kV TX
Contingency: Jajome 3403 138/69 kV Auto Bank 2
Loading: 131.6% (Rating: 145 MVA)
DFAX: 61.6%
Pre-project loading: 84.7%
4. Facility: Elmo – Hickory 138.0 kV L
Contingency: Yarnen – Miravista 345.0 kV Line
Loading: 123.3% (Rating: 323 MVA)
DFAX: 38.6%
Pre-project loading: 110.1%
5. Facility: Ne-Hobbs – No Mans Land 138.0 kV Line
Contingency: Mira Vista – Yarnen 345.0 kV
Loading: 122.2% (Rating: 442 MVA)
DFAX: 38.6%
Pre-project loading: 112.6%

6. Facility: No Maries – Trowbridge 138.0 kV
Contingency: Bel Air – Home 138.0 kV L
Loading: 120.3% (Rating: 235 MVA)
DFAX: 52.3%
Pre-project loading: 95.7%
7. Facility: Jajome 3403 138-69 kV Transformer Bank #2
Contingency: Jajome 3403 138-69 kV Autotransformer Unit 1
Loading: 116.4% (Rating: 164 MVA)
DFAX: 61.6%
Pre-project loading: 74.9%
8. Facility: Home – Bel Air 138.0 kV Line
Contingency: Home – No Maries 138.0 kV
Loading: 114.5% (Rating: 258 MVA)
DFAX: 52.9%
Pre-project loading: 91.8%
9. Facility: Nomaries – Home 138.0 kV
Contingency: Home – Bel Air 138.0 kV
Loading: 113.7% (Rating: 300 MVA)
DFAX: 52.4%
Pre-project loading: 94.4%
10. Facility: Bel Air – Fabrica Tenere 138.0 kV
Contingency: Home Substation – No Aries 138.0 kV L
Loading: 110.1% (Rating: 249 MVA)
DFAX: 52.9%
Pre-project loading: 86.6%
11. Facility: Ne-Hobbs – Harbor Beach 138.0 kV Line
Contingency: Mira Vista – Yarnen 345.0 kV line
Loading: 106.8% (Rating: 534 MVA)
DFAX: 38.6%
Pre-project loading: 98.8%
12. Facility: Harbor Beach – Hickory 138.0 kV
Contingency: Mira Vista – Yarnen 345.0 kV
Loading: 104.9% (Rating: 590 MVA)
DFAX: 38.6%
Pre-project loading: 97.7%
13. Facility: Mitchell Dam – Haggin 345.0 kV
Contingency: Leavitt – Chancellor Row 345.0 kV
Loading: 100.8% (Rating: 1,420 MVA)
DFAX: 19.0%
Pre-project loading: 99.3%
14. Facility: Skaggs Island Jct 2 – Mitchel Dam 345.0 kV L
Contingency: Leavitt – Chancellor Row 345.0 kV
Loading: 100.0% (Rating: 1,754 MVA)
DFAX: 21.4%
Pre-project loading: 98.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Trowbridge 138-69 kV TX	Replace transformer with higher-rated unit	\$3,667,000	41% (\$1,514,000)
Trowbridge 138-69 kV Auto Unit 2	Replace transformer with higher-rated unit	\$4,239,000	41% (\$1,750,000)
Jajome 3403 138/69 kV Auto Unit 1	Replace transformer with higher-rated unit	\$4,466,000	36% (\$1,630,000)
Elmo – Hickory 138.0 kV L	Rebuild line with higher-capacity conductor	\$1,772,000	11% (\$198,000)
Ne-Hobbs – No Mans Land 138.0 kV Line	Rebuild line with higher-capacity conductor	\$4,446,000	10% (\$464,000)
No Maries – Trowbridge 138.0 kV	Rebuild line with higher-capacity conductor	\$14,924,000	41% (\$6,163,000)
Jajome 3403 138-69 kV Transformer Bank #2	Replace transformer with higher-rated unit	\$4,428,000	36% (\$1,616,000)
Home – Bel Air 138.0 kV Line	Rebuild line with higher-capacity conductor	\$20,460,000	41% (\$8,350,000)
Home – No Maries 138.0 kV	Rebuild line with higher-capacity conductor	\$19,936,000	41% (\$8,211,000)
Fabrica Tenere – Bel Air 138.0 kV	Rebuild line with higher-capacity conductor	\$14,302,000	41% (\$5,837,000)
Ne-Hobbs – Harbor Beach 138.0 kV line	Reconductor line	\$2,510,000	10% (\$262,000)
Harbor Beach – Hickory 138.0 kV	Reconductor line	\$2,882,000	10% (\$301,000)
Mitchell Dam – Haggin 345.0 kV	Reconductor line	\$9,271,000	3% (\$253,000)
Skaggs Island Jct 2 – Mitchel Dam 345.0 kV L	Reconductor line	\$20,209,000	3% (\$566,000)
	Total allocated to project		\$37,115,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.